DATA ITEM DESCRIPTION
Title: OVERLOAD AND FAULT ANALYSIS REPORT

Number: DI-SESS-82287
AMSC Number: N10071
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20190815
Limitation: None
GIDEP Applicable: N/A
Project Number: SESS-2019-037

Use/Relationship: The Overload and Fault Analysis Report documents the ability of an electrical power system to safely operate in the presence of overloads and faults. The data will be used to substantiate an airworthiness certification for the system.

This Data Item Description (DID) contains the format, content, and intended use information for the Overload and Fault Analysis Report resulting from fault analysis tasking described by the contract statement of work.

Requirements:

1. Reference Documents. None

2. Format. Contractor format acceptable

3. Content. The report shall include the following:

3.1 All unique overload and fault conditions of the electrical power system

3.2 An electrical bond impedance analysis of the electrical power system and utilization equipment, including:

3.2.1 Bond impedances needed to meet grounding and bonding requirements

3.2.2 Predicted bond impedances for each piece of utilization equipment.

3.2.3 Current carrying capacity of conductors and insulation, including system structural components

3.3 A circuit isolation and fault analysis of the electrical power system, including:

3.3.1 The ability of the system to isolate faults from the sources powering them

3.3.2 The location of each simulated fault and the physical location and electrical connections of the element that interrupted the fault

3.3.3 The ability of the system to operate during and after clearing faults, including

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3.3.3.1 The mission impact of each cleared fault

3.3.3.2 The mission impact of the faults that cannot be cleared

3.3.4 All damage prevention schemes used in the system including:

3.3.4.1 Circuit protection equipment

3.3.4.2 Methods of power regulation and control

3.3.4.3 The ability of each scheme to safely interrupt faults and sustain overloads without creating a fault

3.3.4.4 The response time of damage prevention schemes to overloads and faults

3.3.5 The ability of the system to prevent and isolate are faults for:

3.3.5.1 Steady state voltages

3.3.5.2 Transient voltages

3.3.6 The methods used to prevent the connection of each source to any other source

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